



SPECIFICATION · V1.0

The Query Buckets

The five funnel stages every question is tagged with — what each one measures, how a hundred-query set is split evenly across them, and the three things a bucket decides beyond a label on a chart.

Prepared

5 August 2026

Status

$N = 100 \cdot K = 3$
5 buckets, even split · 20 each

Source of truth

src/prompts/intent.py
data/query_templates.json

Why a bucket, and not just a number

Every query carries exactly one intent tag: where the buyer is standing when they ask it. Without it, an audit returns one undifferentiated figure — “*you appear in 43% of answers*” — which is close to useless. With it, the same run returns a *shape*.

The brand trap, which is what the buckets exist to prevent. Nearly every business scores high on brand queries, because the model has read their website. “*Is {client} any good*” is answered well for almost everyone — and the buyer asking it already knew the name. A single blended percentage is dominated by that number. Bucketing is what stops the most flattering, least valuable stage from becoming the headline.

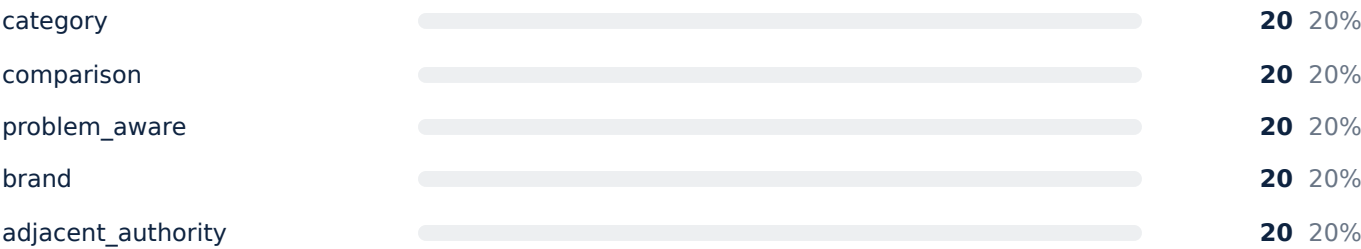
Read by stage instead, the same data says something actionable: “*visible at 100% when asked for by name, 7% when someone describes the problem.*” That is a diagnosis with a next action attached.

The instrument



K = 3 is not redundancy, it is the measurement. These engines are non-deterministic: the same question asked twice returns different answers. One sample of a stochastic system is an anecdote. Three runs makes a mention rate a rate, and is what cycle-over-cycle comparison is computed on.

Allocation at N = 100



The split is even, and that is a measurement decision before it is an editorial one. An unequal allocation makes the funnel chart dishonest: a stage measured on fifteen questions and one measured on thirty carry different confidence, so the gap between them is partly an artefact of sample size. Twenty per bucket — sixty observations each per surface once K = 3 is applied — means every stage is measured to the same precision.

*The consequence: brand takes a full fifth of the instrument, and brand is the stage almost every business already wins. **Never lead with the blended number.** The finding is the gap between stages, not the average of them.*

The five buckets

Ordered cold to warm. Every shape listed is live in the template bank; slots are filled only from a line the client confirmed on the fact sheet or from the session's run inputs, so every generated query traces back to something a human vouched for.

problem_aware

20% of the set

20 queries at N=100

5 SHAPES

Problem-aware

The buyer describes a pain and names neither the category nor any brand.

Least control, most upside. Named here and you have won before anyone is competing. Anchored to real forum and review language; the category must never slip in.

COMPOSITION RULE First person. No category, no brand.

TEMPLATE SHAPES · 5 IN THE BANK

4 NEAR VERBATIM

1 CONSTRUCTED

- my {category} keeps breaking, what should I do
- how do I know when I need a {category}
- is it worth paying for {category} or should I do it myself
- what does it usually cost to fix this myself vs hiring someone
- how do I choose between {category} options without getting ripped off

category

20% of the set

20 queries at N=100

9 SHAPES

Category / solution-aware

They know what they need. They do not know who provides it.

Where the buying decision is actually formed. One head query; every other carries a qualifier drawn from a confirmed fact-sheet line. Several are year-stamped, to bait staleness against the current version.

COMPOSITION RULE Category named. Client not.

TEMPLATE SHAPES · 9 IN THE BANK

2 VERBATIM

6 NEAR VERBATIM

1 CONSTRUCTED

- best {category}
- best {category} in {city}
- most reliable {category} for someone on a budget
- top rated {category} {year}
- who are the leading {category} providers right now
- {category} recommendations for a first-time buyer
- what should I look for in a {category}
- cheapest decent {category}
- {category} near {city} that people actually recommend

comparison

20% of the set

20 queries at N=100

5 SHAPES

Comparison

Head-to-head against a named rival.

Late and high intent. Two hard constraints live here: every competitor must appear at least once, and at least two queries must leave the client unnamed — those test unprompted surfacing, which is the measurement that matters most.

COMPOSITION RULE At least one competitor named.

TEMPLATE SHAPES · 5 IN THE BANK

2 VERBATIM

3 NEAR VERBATIM

- {client} vs {competitor}
- is {client} better than {competitor}
- best alternative to {competitor}
- {competitor} alternatives that cost less
- how does {client} compare to {competitor} on price

brand

20% of the set

20 queries at N=100

6 SHAPES

Brand / bottom-funnel

They already know the name and are checking it out.

The flattering number. Probes the claims most damaging if wrong — price, mandatory fees, whether the company is legitimate. This is the bucket where accuracy flags surface, which is why it earns a full share despite being the easiest stage to win.

COMPOSITION RULE Client named. The only bucket where that is allowed.

TEMPLATE SHAPES · 6 IN THE BANK

4 VERBATIM

2 NEAR VERBATIM

- is {client} any good
- how much does {client} cost
- {client} reviews
- is {client} legit
- what does {client} actually offer
- does {client} have any hidden fees

adjacent_authority

20% of the set

20 queries at N=100

5 SHAPES

Adjacent authority

Expertise topics in the client's domain, with no brand named at all.

Tests whether the business reads as the expert rather than a listing. The topic must map to expertise the client could plausibly own.

COMPOSITION RULE No brand named, by anyone.

TEMPLATE SHAPES · 5 IN THE BANK

4 NEAR VERBATIM

1 CONSTRUCTED

- how does {category} pricing usually work
- what questions should I ask before hiring a {category}
- common mistakes people make with {category}
- how long should a {category} job take
- signs you are being overcharged for {category}

What a bucket decides

A bucket is not a label on a chart. It controls three things.

One — the reporting axis

Mention rate, prominence and framing are all reported per bucket. The headline finding of most audits is not the blended number but the *gap* between two stages — strong on brand, absent on category. That gap is the recommendation.

Two — engine routing

The run is a uniform `N × engines × K` cross product, which assumes every surface can answer every kind of question. A routing layer exists for the cases where one cannot: a policy names the intents an engine should be skipped for, and those cells are never spent.

No policy currently applies to this funnel. The one shipped policy is a denylist against an intent that consumer sets do not contain, so a consumer run is the full cross product and the routing layer is inert by construction — pinned by a regression test rather than left to be remembered. The layer matters here as headroom: when a surface is measured to be structurally absent for one stage, it can be switched off for that stage alone without touching the others.

Three — set composition and the QA gate

Bucket membership is what the pre-run lint checks against. Blocking failures stop a set being approved; warnings are surfaced for a human to overrule.

CHECK	LEVEL	WHY
Exactly one valid intent per query	block	The CSV parser rejects the file.
Every competitor appears in ≥ 1 comparison query	block	A competitor nobody asks about is a competitor nobody measured.
≥ 2 comparison queries leave the client unnamed	block	Tests unprompted surfacing.
No unfilled <code>{slot}</code> survives	block	Scores as a loss on a question nobody asked.
Client named only in brand and client-named comparisons	warn	Naming the client leads the answer.
Counts within $\pm 20\%$ of the allocation	warn	Balance, not law.
$\geq 1/3$ of the set verbatim or near-verbatim	warn	Guards against language nobody uses.
Near-duplicate sweep	warn	Two phrasings measure one thing twice.

The template bank

Thirty shapes across the five buckets. Shapes, not questions: every slot is filled from a confirmed fact-sheet line or a session run input, so nothing reaches an engine that a human did not vouch for.

No model writes an entry in this bank. The methodology forbids LLM-originated queries — the drafter's role is *source, draft, format*, never originate from imagination. Adding a shape is a human editorial act. A generator that invented its own questions would be measuring language no buyer uses, and would do it non-deterministically in a screen whose entire job is human review.

Provenance, recorded per shape

TAG	MEANING	IN THE BANK
verbatim	Lifted from a real buyer — a review, a ticket, a call transcript.	8
near_verbatim	A real phrasing with the brand swapped out.	19
constructed	A pure shape nobody said in these words.	3

A shape whose slots cannot all be filled is dropped, never partially filled. A half-rendered query is worse than a missing one: it reaches a live engine, measures a question nobody asked, and scores as a loss.

The bank has to grow to reach N = 100

Thirty shapes cannot produce a hundred distinct queries on their own. Comparison multiplies cleanly — five shapes across four named competitors is twenty real queries. The other four buckets do not: they have no per-item slot to multiply across, so at N = 100 they would reuse each shape two to four times and trip the near-duplicate sweep.

BUCKET	NEEDED AT N = 100	SHAPES IN BANK	REUSE	RESOLUTION
comparison	20	5	×4 across competitors	Fine — multiplies on the competitor list
category	20	9	×2.2	Add a qualifier slot filled from the offerings
brand	20	6	×3.3	Add ~14 sourced shapes
problem_aware	20	5	×4	Add ~15 sourced shapes
adjacent_authority	20	5	×4	Add a topic slot filled from the offerings

The cheapest route is not forty more hand-written shapes. It is a `{qualifier}` slot on the category and adjacent-authority shapes, filled from the offerings the client confirmed on the fact sheet — *best {category} for {qualifier}* across eight named offerings is twenty distinct, sourced, non-duplicate queries from two shapes. That also keeps the guarantee this bank exists to make: every query traces to a line a human vouched for.

What a cycle costs

PATH	N	ENGINES	K	CALLS
Standard instrument	100	5	3	1,500
Calibration trim	15-18	5	3	225-270

The dollar figure depends on the engine mix and is estimated per set at review time. Judging runs against a cache keyed on the fact sheet, so re-judging an unchanged sheet is free — but approving a new sheet version re-keys every cached verdict for that client, and the next cycle re-judges.